LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	15
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Motor actuators: - Internal actuators, rotary actuators, actuator gearing, limit s/w, breaks, clutches.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Motor actuators: - Internal actuators, rotary actuators, actuator gearing, limit s/w, breaks, clutches.
5. Specific Objectives	13.1.3.1 Knowledge on circuit protection devices, generators, actuators & engine starting system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of actuator schematics in IB; verify limit switch positions on diagrams; instructor supervises diagram tracing.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a motor actuator?
(b) Name two types of actuators covered in this syllabus.
(c) Why are limit switches critical in actuator systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Internal actuators construction • Rotary actuators operation • Actuator gearing mechanisms • Limit switch safety functions • Brake and clutch integration	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of internal actuator components • Functionality of rotary actuator gearing • Operational limits and safety protocols • Role of brakes in actuator systems • Clutch engagement and disengagement	L & Ds	PPT, PDF	20 min
(c)	• Comparison of internal vs rotary actuator applications • Troubleshooting common actuator failures • Maintenance requirements for gears and clutches • Safety checks for limit switches • Integration with engine starting systems	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between internal and rotary actuators.
(b) Describe the function of limit switches in preventing over-travel.
(c) How do brakes and clutches contribute to actuator safety?

12. Home Assignment and Follow Up	Prepare a brief note on the purpose and operation of A/C Internal lighting, focusing on instrument illumination lamps and control panel lights, to be discussed in LP 16.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Purpose & operation of A/C Internal lighting: - Instrument illumination lamp, control panel light, cabin lights, map lights, emergency escape light passenger information light. Practical on position & wiring of lights.	Refer to the master syllabus reference list	LP 16

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min